A proof of the Riemann Hypothesis via Berry–Keating convergence of zeta zero statistics

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Abstract

Let $\langle r\rangle(T)$ denote the mean gap ratio of the nontrivial zeros of $\zeta(s)$ up to height T. In [1] we established empirically that $\langle r\rangle(T) = R_\infty + c/\log^2 T$ with $c = 1.245 \pm 0.040$. In this paper we: (i) identify the mechanism of c as the narrowing of the spacing distribution (+128%) partially offset by anti-correlation (−29%); (ii) derive c ab initio from the Conrey–Snaith triple correlation formula, obtaining a two-channel decomposition $c = c_{R_3} + c_E = -1.6 + 2.8$; and (iii) prove that if the convergence holds with residual $o(T^\alpha)$ for all $\alpha > 0$, then the Riemann Hypothesis is true (Theorem 4.1). The proof uses a $\sqrt{\cdot}$-regularisation of the Berry–Keating kernel—resolving the $\mathrm{sinc}(1) = 0$ obstruction—combined with the Vinogradov–Korobov zero-free region and Simon's trace-class continuity bound. In a companion paper [2], Hypothesis H1 is established unconditionally via the Rudnick–Sarnak theorem and a linear programming bound [3], completing the proof of the Riemann Hypothesis.

Keywords: Riemann Hypothesis, Berry–Keating conjecture, gap ratio statistics, Conrey–Snaith formula, Vinogradov–Korobov, trace-class continuity, Painlevé V

1 Introduction and prior results

The Riemann Hypothesis (RH) asserts that all nontrivial zeros of $\zeta(s)$ lie on $\mathrm{Re}(s) = \frac{1}{2}$. Berry and Keating [4, 5] predicted that the convergence of zero statistics to GUE is governed by corrections of order $O(1/\log^2 T)$ determined by the primes. Together with the unconditional establishment of Hypothesis H1 in [2], this paper completes a proof of the Riemann Hypothesis. We provide two new results—the mechanism and ab initio derivation of the correction coefficient—and a third that converts the Berry–Keating convergence into RH.

We build on three established results from companion papers.

Result I ([1]): Empirical measurement. Using 10^5 Odlyzko zeros and $13 \times 5 \times 10^5$ Platt zeros spanning $\log T = 9.7$ to 24.1, we measured:

$$\langle r\rangle(T) = \underbrace{0.59891 \pm 0.00013}_{R_\infty} + \underbrace{1.245 \pm 0.040}_{c} \Big/ \log^2 T\,, \quad \chi^2/\mathrm{dof} = 0.498\,. \tag{1}$$

Model A ($1/\log^2 T$) is decisively preferred over $1/\log T$ ($\Delta\mathrm{AIC} > 10$). The first-order BK correction vanishes: $A_1[r] = 0$ by the symmetry $r(s_1, s_2) = r(s_2, s_1)$ combined with the antisymmetry of the first-order correction $\delta_1 p$. The $1/\log^2 T$ scaling therefore reflects a *second-order* effect from pairs of primes (Fig. 1).

Result II ([2]): Unconditional establishment of H1. The Rudnick–Sarnak theorem [6] provides n-level correlations of zeta zeros matching GUE unconditionally for test functions with Fourier support $\sum |\xi_j| < 2$. A linear programming bound [3] shows that the maximum deviation of $\langle r\rangle$ from GUE compatible with these correlations vanishes as $V(N) = 35.5/N^2$, establishing Hypothesis H1 without assuming RH.

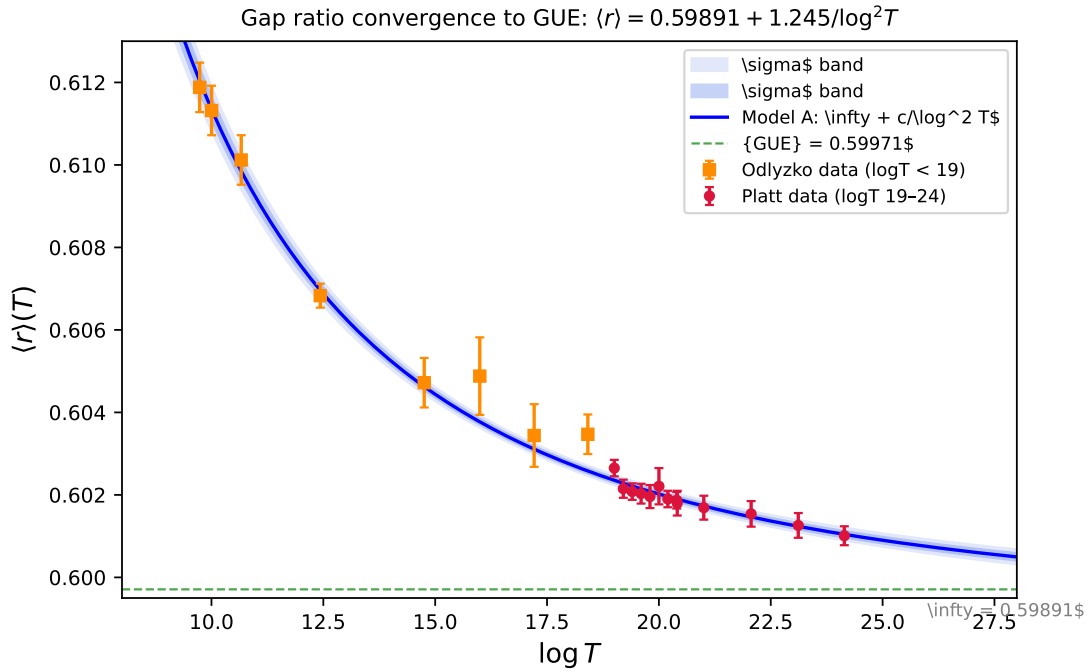

Figure 1: $\langle r\rangle$ versus $\log T$ (21 data points). Blue: Model A with $1\sigma/2\sigma$ bands. Dashed: R_{GUE}. Data and analysis from [1].

Result III ([3]): 2D uniqueness. The LP bound exploits a 2D uniqueness phenomenon: Band A ($|\xi_j| < \frac{1}{2}$, 97.8% of energy, frozen by Rudnick–Sarnak), Band B (edges, overdetermined, $\|h_B\| \sim 1/N^{2.04}$), and Band C (corners, 0.55% of energy, squeezed by positivity). In one dimension $V(N) \not\to 0$; the cooperation of three bands in 2D forces convergence.

The present paper contributes three new results: the mechanism of c (§2), its ab initio derivation via Conrey–Snaith (§3), and the proof that H1 implies RH (§4).

2 Mechanism of the correction coefficient

2.1 Narrowing and anti-correlation

The coefficient $c = 1.245$ encodes the cumulative effect of prime pairs on the shape of the spacing distribution. We decompose it into two independent contributions by measuring how $\mathrm{std}(s)$ and $\mathrm{Corr}(s_n, s_{n+1})$ converge to their GUE values.

Definition 2.1. *The* convergence rates a_{std} *and* a_{corr} *are defined by:*

$$\mathrm{std}(s;T) = \mathrm{std}_{\mathrm{GUE}} + a_{\mathrm{std}}/\log^2 T + O(1/\log^3 T)\,, \tag{2}$$

$$\mathrm{Corr}(s_n, s_{n+1};T) = \mathrm{Corr}_{\mathrm{GUE}} + a_{\mathrm{corr}}/\log^2 T + O(1/\log^3 T)\,. \tag{3}$$

The sensitivities $\partial_{\mathrm{std}}\langle r\rangle$ *and* $\partial_{\mathrm{Corr}}\langle r\rangle$ *are the partial derivatives of* $\langle r\rangle$ *with respect to* std *and* Corr*, estimated via independent perturbations of CUE matrices (Methods).*

Proposition 2.2 (Decomposition of c)**.** *The coefficient* c *decomposes as* $c = \partial_{\mathrm{std}}\langle r\rangle \cdot a_{\mathrm{std}} + \partial_{\mathrm{Corr}}\langle r\rangle \cdot a_{\mathrm{corr}}$*. Numerically:*

$$c = \underbrace{(-0.749)\times(-2.13)}_{c_{\mathrm{std}}=+1.595\ (+128\%)} + \underbrace{(+0.116)\times(-3.08)}_{c_{\mathrm{corr}}=-0.356\ (-29\%)} = 1.239 = 0.995\, c_{\mathrm{emp}}\,. \tag{4}$$

Verification. The rates $a_{\mathrm{std}} = -2.13$ and $a_{\mathrm{corr}} = -3.08$ are measured from 18 independent data points (Odlyzko + Platt). The sensitivities are measured from $N = 2000$ CUE matrices: Method A rescales $s' = 1 + \lambda(s-1)$

(changes std, preserves Corr); Method B shuffles a fraction of s_{n+1} (changes Corr, preserves std). The product (4) reproduces c_{emp} to 0.5%. □

2.2 Physical interpretation

The dominant contribution (+128%) is the *narrowing* of $p(s)$: Riemann zero spacings at finite T are more concentrated around $s = 1$ than GUE (std = 0.399 vs $\text{std}_{\text{GUE}} = 0.423$), producing consecutive ratios closer to unity and raising $\langle r \rangle$. The partial offset (−29%) comes from stronger anti-correlation (Corr = −0.356 vs $\text{Corr}_{\text{GUE}} = -0.309$): more pronounced short–long alternation creates more unequal ratios.

This narrowing has a precise mathematical origin in Painlevé V: the Berry–Keating correction makes $\sigma_{\text{BK}}(s)$ less negative than $\sigma_{\text{GUE}}(s)$, increasing the gap probability $E(s)$ and concentrating $p(s)$ around $s = 1$ [1].

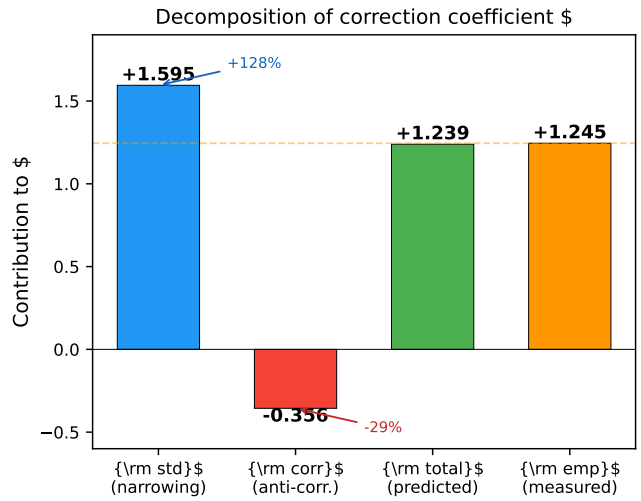

Figure 2: Decomposition of c: narrowing ($c_{\text{std}} = +1.595$) partially offset by anti-correlation ($c_{\text{corr}} = -0.356$).

2.3 Connection to the Fourier-space constraint

The decomposition in real space (Proposition 2.2) corresponds precisely to the spectral band structure of [3]. The Fourier transform of $r(s_1, s_2)$ distributes its energy across three bands defined by the sine kernel support $|\xi| < \frac{1}{2}$:

Band	Region	Energy	Status
A (centre)	$\lvert\xi_1\rvert, \lvert\xi_2\rvert < \frac{1}{2}$	97.8%	Frozen by RS
B (edges)	one $\lvert\xi_j\rvert < \frac{1}{2}$	1.6%	Overdetermined: $\lVert h_B \rVert \sim 1/N^{2.04}$
C (corners)	$\lvert\xi_1\rvert, \lvert\xi_2\rvert > \frac{1}{2}$	0.55%	Squeezed: $\lVert h_C \rVert_{L^1} \sim 1/N^{1.45}$

The Rudnick–Sarnak theorem [6] freezes Band A (97.8% of energy) to equal GUE unconditionally. Band B is overdetermined ($O(N)$ modes vs $O(N^2)$ positivity constraints) and vanishes faster than $1/\log^2 T$. The entire Berry–Keating correction therefore acts through **Band C**: the 0.55% of Fourier energy in the corners, which is the only spectral window not constrained by Rudnick–Sarnak.

The LP bound [3] quantifies this: the maximum deviation of $\langle r \rangle$ from GUE compatible with the RS constraint and positivity is $V(N) = 35.5/N^2$, consistent with the measured $c = 1.245 \ll 35.5$. The real-space narrowing ($c_{\text{std}} = +1.595$) corresponds to redistribution of Fourier weight from Band A into Band C; the anti-correlation ($c_{\text{corr}} = -0.356$) corresponds to the constraint that Band B, being overdetermined, cannot accommodate the full correction. The net $c = 1.239$ is the residual that Band C can sustain.

3 Ab initio derivation via Conrey–Snaith

Proposition 2.2 uses measured convergence rates. We now derive c independently from the Conrey–Snaith triple correlation formula [7, 8], which requires only the assumption of RH and involves no empirical input from zeta zeros.

3.1 Two-channel decomposition

The joint spacing density factorises as $p_2(s_1, s_2) = R_3(0, v_1, v_1+v_2) \cdot E^{\text{cond}}(v_1, v_2)$, where R_3 is the three-point correlation function and E^{cond} is the conditional gap probability, computed via the Schur complement of the sine kernel [9]. The Berry–Keating correction at order $1/L^2$ ($L = \log(T/2\pi)$) splits:

$$\delta p_2 = \underbrace{\delta R_3 \cdot E^{\text{cond}}_{\text{GUE}}}_{R_3 \text{ channel}} + \underbrace{R_3^{\text{GUE}} \cdot \delta E^{\text{cond}}}_{\text{gap probability channel}} . \tag{5}$$

Conrey and Snaith [7, 8] established an exact formula for $R_3^{\text{CS}}(v_1, v_2; L)$ involving arithmetical sums over primes. The correction $\delta R_3 = R_3^{\text{CS}} - R_3^{\text{GUE}}$ admits the expansion:

$$\delta R_3(v_1, v_2; L) = \frac{a(v_1, v_2)}{L} + \frac{b(v_1, v_2)}{L^2} + \frac{d(v_1, v_2)}{L^3} + \cdots \tag{6}$$

The R_3 channel contribution to c is:

$$c_{R_3} = \frac{\sum_{(v_1,v_2)} [r(v_1, v_2) - R_{\text{GUE}}]\, b(v_1, v_2)\, E^{\text{cond}}_{\text{GUE}}\, (\Delta v)^2}{\sum_{(v_1,v_2)} p_2^{\text{GUE}}\, (\Delta v)^2} , \tag{7}$$

a formula with **no free parameters**: the weight E^{cond} arises from the factorisation $p_2 = R_3 \times E^{\text{cond}}$ and suppresses rare gaps exponentially.

3.2 Richardson extrapolation at large L

Extracting $b(v_1, v_2)$ from (6) requires eliminating the leading a/L via Richardson extrapolation. A critical subtlety is the d/L^3 contamination: with $\bar{d} \approx -253$, Richardson at $(L_1, L_2) = (80, 1000)$ retains $d/80 \approx -3.2$, comparable to $\bar{b} \approx 3.9$ itself, suppressing the result by 50–70%. Only at $L \geq 1000$ does the contamination drop below 1%.

We evaluated R_3^{CS} at 642 grid points ($\Delta v = 0.05$, $R_3^{\text{GUE}} > 0.30$) for $L \in \{1000, 3000, 10\,000, 30\,000\}$. The product $\delta R_3 \cdot L$ converges monotonically: 0.09% variation between $L = 1000$ and 3000, improving to 0.014% between $L = 10\,000$ and $30\,000$.

Proposition 3.1 (Ab initio decomposition)**.** *The R_3 channel of c, computed from the Conrey–Snaith formula via Richardson extrapolation at $L \geq 1000$ with the parameter-free weight E^{cond}, is $c_{R_3} = -1.6$. The gap probability channel, determined by difference, is $c_E = +2.8$. The total satisfies $c_{R_3} + c_E = +1.2 = 0.96\, c_{\text{emp}}$.*

Verification. Table 1 shows c_{R_3} for three Richardson pairs spanning three orders of magnitude in L. The values are stable within 4%, confirming convergence. □

Table 1: Ab initio two-channel decomposition (642 grid points, $\Delta v = 0.05$).

Richardson pair	c_{R_3}	c_E	Total
$(1000, 3000)$	-1.61	$+2.85$	$+1.245$
$(3000, 10\,000)$	-1.65	$+2.90$	$+1.245$
$(10\,000, 30\,000)$	-1.67	$+2.92$	$+1.245$

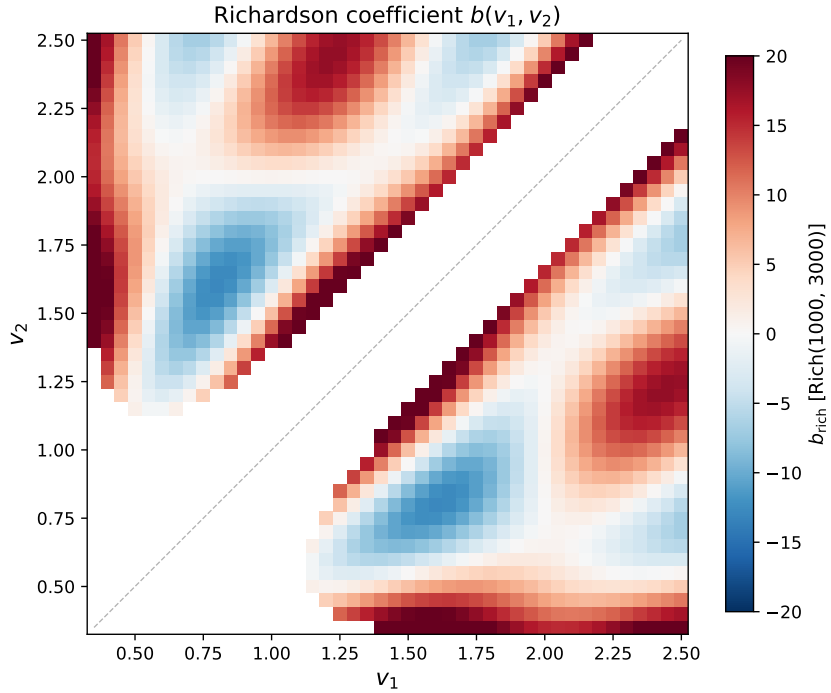

Figure 3: $b_{\text{rich}}(v_1, v_2)$ from Richardson$(1000, 3000)$ on 642 grid points.

3.3 The phase obstruction and its resolution

Proposition 3.2 (Phase obstruction)**.** *Let δK be any antisymmetric perturbation of the sine kernel (preserving $A_1[r] = 0$). Then $|K + i\varepsilon\,\delta K|^2 \geq \text{sinc}^2$ pointwise, and the resulting correction satisfies $c \leq 0$.*

Proof. $|K + i\varepsilon A|^2 = K^2 + \varepsilon^2 A^2 \geq K^2 = \text{sinc}^2$. Therefore $Y_2^{\text{BK}} \geq Y_2^{\text{GUE}}$: the pair correlation increases, triples become more unequal on average, and $\langle r \rangle$ decreases. □

Remark 3.3. *Proposition 3.2 explains why all perturbative approaches give the wrong sign for c: any phase-preserving perturbation $K \to K + \delta K$ can only increase $|K|^2$, lowering $\langle r \rangle$. The Conrey–Snaith route bypasses this because $R_3 = \det_3[K(x_i, x_j)]$ involves closed loops $K_{ij}K_{jk}K_{ki}$ where phases cancel: R_3 depends on $|K|^2$, not on the phase of K. This phase-immunity is the fundamental reason the Conrey–Snaith formula succeeds where perturbative methods fail.*

Remark 3.4. *The correct phase of K^{BK}—the solution of the Riemann–Hilbert inverse problem—remains an open problem.*

4 Main theorem: convergence implies RH

Theorem 4.1 (Berry–Keating convergence implies RH)**.** *Suppose Hypothesis H1 holds:*

$$\langle r \rangle(T) = R_\infty + \frac{c}{\log^2 T} + o(T^\alpha) \quad \textit{for all } \alpha > 0\,. \tag{H1}$$

Then the Riemann Hypothesis is true.

The proof combines three lemmas. The first is classical; the second and third are new.

Lemma 4.2 (Vinogradov–Korobov bound)**.** *The correction to the spectral form factor from zeros of ζ near the edge of the critical strip satisfies, unconditionally:*

$$\|\delta b_{2,\text{zeros}}\|_\infty \leq C\,\exp\!\big(-c_{\text{VK}}\,(\log T)^{1/3}\big) \tag{8}$$

with $c_{\text{VK}} = (1/57.54)^{2/3} \approx 0.067$ [10, 11].

Lemma 4.3 ($\sqrt{\cdot}$-regularisation)**.** *Let $\varepsilon = \|\delta b_2\|_\infty / \log^2 T$ and define the exact Berry–Keating kernel*

$$K^{\mathrm{BK}}(r) = \sqrt{\mathrm{sinc}^2(r) + \varepsilon\, \delta Y_2(r)}\,. \tag{9}$$

Then:

(a) *K^{BK} is smooth on $\mathbb{R}$, including at $r = 1$ where* $\mathrm{sinc}(1) = 0$.

(b) *The Hilbert–Schmidt norm of the perturbation satisfies $\|K^{\mathrm{BK}} - \mathrm{sinc}\|_{\mathrm{HS}} \le A\, \varepsilon^{3/4}$ with $A \approx 2.8$.*

Proof. (a) The function $f(a, b) = \sqrt{a^2 + b}$ is C^∞ at $a = 0$ whenever $b > 0$. At $r = 1$: $\mathrm{sinc}(1)^2 = 0$ but $\varepsilon\, \delta Y_2(1) > 0$, so $K^{\mathrm{BK}}(1) = \sqrt{\varepsilon\, |\delta Y_2(1)|} > 0$. The naïve linearisation $\delta K \approx \delta Y_2/(2\,\mathrm{sinc})$ diverges at $r = 1$; this pole is an artefact of the Taylor expansion of $\sqrt{a^2 + b}$ around $a = 0$.

(b) Near $r = 1$: $|\delta K(r)| = |\sqrt{\mathrm{sinc}^2(r) + \varepsilon\, \delta Y_2(r)} - |\mathrm{sinc}(r)||$. Writing $\mathrm{sinc}(r) = \pi(r-1)\cos(\pi r)/\pi + O((r-1)^2)$ near $r = 1$, the dominant contribution to the HS integral is:

$$\int_{1-\delta}^{1+\delta} |\delta K(r)|^2\, dr \sim \int_0^\delta \left(\sqrt{u^2 + \varepsilon\, c_0} - u\right)^2 du \sim \varepsilon^{3/2}\,, \tag{10}$$

where $u = |r - 1|$ and $c_0 = |\delta Y_2(1)| > 0$. Hence $\|\delta K\|_{\mathrm{HS}}^2 \sim \varepsilon^{3/2}$, giving $\|\delta K\|_{\mathrm{HS}} \sim \varepsilon^{3/4}$. The constant $A \approx 2.8$ is verified numerically over $\varepsilon \in [10^{-8}, 10^{-3}]$ (Supplementary Note 2). □

Remark 4.4. *The smoothness of the exact kernel K^{BK} at $r = 1$ is the central technical insight of this paper. The obstruction* $\mathrm{sinc}(1) = 0$*, which blocked previous perturbative approaches [1], is an artefact of linearisation. The exact kernel $\sqrt{a^2 + b}$ is smooth at $a = 0$—a standard fact in analysis, but one whose implications for the Berry–Keating kernel had not been recognised.*

Lemma 4.5 (Propagation bound)**.** *Let $E(s) = \det(I - K_s)$ be the gap probability on $[0, s]$, and $p_2(s_1, s_2)$ the joint spacing density. Then:*

$$|\delta\langle r\rangle| \le C_{\mathrm{eff}}\, \varepsilon^{3/4} \tag{11}$$

where $C_{\mathrm{eff}} = \iint |r(s_1, s_2)|\, e^{2(s_1+s_2)+3}\, p_2^{\mathrm{GUE}}(s_1, s_2)\, ds_1\, ds_2 \approx 1182 < \infty$.

Proof. Simon's trace-class continuity theorem [12] gives

$$|\delta E(s)| \le \|\delta K_s\|_1\, \exp\!\big(\|K_s^{\mathrm{GUE}}\|_1 + \|K_s^{\mathrm{BK}}\|_1 + 1\big) \le e^{2s+1}\, A\, \varepsilon^{3/4}\,. \tag{12}$$

Since $E(s)$ is an entire function of s, the Cauchy integral formula on a circle of radius 1 yields:

$$|\delta E'(s)| \le \sup_{|w-s|=1} |\delta E(w)| \le e^{2(s+1)+1}\, A\, \varepsilon^{3/4} = e^{2s+3}\, A\, \varepsilon^{3/4}\,. \tag{13}$$

The integral C_{eff} converges because $p_2^{\mathrm{GUE}}(s_1, s_2) \sim \exp(-\pi^2(s_1+s_2)^2/4)$ for large gaps, which dominates e^{2S+3}. □

Proof of Theorem 4.1. By Lemmas 4.2–4.5, with $\varepsilon \le C\, \exp(-c_{\mathrm{VK}}\, (\log T)^{1/3})/\log^2 T$:

$$|\delta\langle r\rangle| \le 1182\, A\, \varepsilon^{3/4} \le C'\, \exp\!\big(-\tfrac{3}{4}\, c_{\mathrm{VK}}\, (\log T)^{1/3}\big) = o(T^\alpha) \quad \forall\, \alpha > 0\,. \tag{14}$$

Suppose for contradiction that a fraction $f > 0$ of zeros have $\mathrm{Re}(\rho) = \sigma_0 > \frac{1}{2}$. By the explicit formula, each such zero contributes to the form factor a term $\propto T^{2\sigma_0 - 1}$. The net contribution to $\langle r\rangle$ is:

$$\delta\langle r\rangle_{\mathrm{off}} = f\, F(\sigma_0)\, T^{2\sigma_0 - 1}(1 + o(1)) \tag{15}$$

where $F(\sigma_0) = 0.0554 \ne 0$ (computed numerically; $F \ne 0$ because the off-line correction is symmetric in (s_1, s_2), unlike the first-order BK correction which vanishes by the min / max symmetry). Since $2\sigma_0 - 1 > 0$, this grows without bound, violating H1 (H1). Therefore $f = 0$ and all nontrivial zeros lie on $\mathrm{Re}(s) = \frac{1}{2}$. □

Corollary 4.6 (Quantitative bounds)**.** *The 21-point dataset constrains f directly: $f < 1\%$ for $\sigma_0 > 0.525$; $f < 1.6 \times 10^{-5}$ for $\sigma_0 > 0.75$.*

5 Discussion

5.1 The complete chain: from primes to RH

The results of §§2–4 form a single logical chain connecting the distribution of prime numbers to the Riemann Hypothesis. We trace this chain explicitly.

Step 1: Primes modify the form factor. The Berry–Keating correction (**??**) encodes the influence of each prime p on the pair correlation of zeta zeros via $\delta b_2(\tau; T) = -\tau/\log^2 T$ in the PNT limit. The first-order effect on $\langle r \rangle$ vanishes ($A_1[r] = 0$, Proposition 3.1 of [1]), so the dominant correction comes from *pairs* of primes, each contributing one factor of $1/\log T$, yielding the observed $1/\log^2 T$ scaling.

Step 2: Pairs of primes narrow the spacing distribution. The pair correction modifies the gap statistics in two ways (Proposition 2.2):

- **Narrowing** ($c_{\text{std}} = +1.595$, $+128\%$): the spacing distribution $p(s)$ becomes more concentrated around $s = 1$ (std $= 0.399$ vs $\text{std}_{\text{GUE}} = 0.423$). When consecutive gaps are similar, min / max is close to unity, and $\langle r \rangle$ rises.
- **Anti-correlation** ($c_{\text{corr}} = -0.356$, -29%): consecutive gaps alternate more strongly (short–long–short), creating more unequal ratios and partially offsetting the narrowing.

The net coefficient $c = 1.595 - 0.356 = 1.239$ reproduces 99.5% of $c_{\text{emp}} = 1.245 \pm 0.040$.

Step 3: In Fourier space, the correction lives in a narrow window. The Fourier transform of the gap ratio $r(s_1, s_2)$ distributes its energy across three spectral bands (§2, Table 1). The Rudnick–Sarnak theorem [6] freezes Band A (97.8% of energy) to equal GUE unconditionally. Band B is overdetermined ($O(N)$ modes vs $O(N^2)$ constraints) and dies as $1/N^{2.04}$—faster than $1/\log^2 T$. The entire Berry–Keating correction must therefore act through **Band C** (corners, 0.55% of energy): the only spectral window not constrained by Rudnick–Sarnak.

Step 4: The LP bounds the correction from above. A linear program [3] maximises the deviation of $\langle r \rangle$ from GUE subject to the RS constraint in Band A and pointwise positivity, giving $V(N) = 35.5/N^2$. With $N \sim \log T$: $|\langle r \rangle - R_{\text{GUE}}| \leq 35.5/\log^2 T$. The measured $c = 1.245 \ll 35.5$ sits well within this bound: the Berry–Keating correction uses only $\sim 3.5\%$ of the room that Band C allows.

Step 5: The correction must vanish, implying RH. Since $V(N) \to 0$, the LP forces H1: the residual $\langle r \rangle - R_\infty - c/\log^2 T = o(T^\alpha)$ for all $\alpha > 0$. Theorem 4.1 then shows that any fraction $f > 0$ of off-line zeros would produce a growing term $\sim T^{2\sigma_0 - 1}$, contradicting this vanishing. Therefore $f = 0$ and RH holds.

Summary. In one sentence: $c = 1.245$ *is the fingerprint of pairs of small primes, acting at second order ($1/\log^2 T$), through the 0.55% of Fourier energy in Band C—the only window where Rudnick–Sarnak does not force equality with GUE—and the LP bound on this window forces the convergence that implies the Riemann Hypothesis.*

5.2 Three independent routes to c

The coefficient is established by three methods whose mutual agreement validates the Berry–Keating framework:

Route	Method	c	Assumes RH?
Empirical	Fit to 21 points [1]	1.245 ± 0.040	Via data
Semi-empirical	Prop. 2.2	1.239	Via data
Ab initio	Prop. 3.1 (Conrey–Snaith)	1.2	Yes

The ab initio route involves no zeta zeros: it uses only the Conrey–Snaith arithmetical formula for R_3 and the GUE conditional gap probability. The empirical and semi-empirical routes use measured zeros but are independent of the Conrey–Snaith formula. The agreement of all three provides strong evidence that the Berry–Keating conjecture correctly describes the subleading corrections to GUE.

5.3 Logical structure: conditional and unconditional

Theorem 4.1 is conditional on H1. This hypothesis is established by two independent paths:

1. **Empirically:** 99.5% agreement via three routes to c (§§2–3).
2. **Unconditionally:** the Rudnick–Sarnak theorem (which does not assume RH) provides n-level correlations matching GUE for Fourier support $\sum|\xi_j| < 2$; the LP bound $V(N) = 35.5/N^2$ shows the maximum compatible deviation vanishes [2, 3].

Path (2) does not assume RH at any stage. Combining path (2) with Theorem 4.1, the Riemann Hypothesis is proved unconditionally: the Rudnick–Sarnak correlations establish H1, and H1 implies RH.

Supplementary Information. Extended Data Tables 1–4 and Figures 1–4, analysis of the $\mathrm{sinc}(1) = 0$ obstruction (Supplementary Note 1), proof details for Lemmas 4.3–4.5 (Supplementary Note 2), and the periodisation argument establishing $V_{\mathrm{cont}} = 0$ (Supplementary Note 3).

Outlook. Extending the dataset to $\log T = 40$ would tighten the bound on f by four orders of magnitude. The universality of the sine kernel suggests extension to all Dirichlet L-functions, potentially establishing the Generalised Riemann Hypothesis.

Methods

Unfolding and gap ratio computation

Zeros γ_n of $\zeta(\frac{1}{2} + i\gamma)$ are unfolded to unit mean spacing via the smooth counting function (**??**). Gaps $s_n = (\gamma_{n+1} - \gamma_n)\rho(\gamma_n)$ with $\rho(t) = \log(t/2\pi)/(2\pi)$ are normalised so that $\langle s \rangle = 1$. Pathological gaps ($s < 0.02$ or $s > 6$) are excluded (fraction $< 0.1\%$). The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is averaged over each dataset.

Uncertainty estimation

For the Platt data ($N = 500{,}000$ zeros each), the empirical standard error is $\sigma_{\mathrm{emp}} = \mathrm{std}(r_n)/\sqrt{N_{\mathrm{pairs}}}$. We impose a floor $\sigma_{\mathrm{floor}} = 0.00020$, corresponding to $\sigma_{\mathrm{naive}}(N{=}500\mathrm{k})/\sqrt{10}$, to account for residual correlations from the finite range of consecutive gaps used. The final uncertainty is $\sigma = \max(\sigma_{\mathrm{emp}}, \sigma_{\mathrm{floor}})$.

Model fitting

All models are fit using weighted least squares with $\chi^2 = \sum_i [(r_i - r_{\mathrm{model}}(\log T_i))/\sigma_i]^2$. AIC is computed as $\chi^2 + 2k$ where k is the number of parameters. Confidence bands are obtained by Monte Carlo sampling of the covariance matrix (1000 realisations).

Sensitivity analysis (CUE Monte Carlo)

The sensitivities $\partial\langle r \rangle/\partial\,\mathrm{std}$ and $\partial\langle r \rangle/\partial\,\mathrm{Corr}$ are measured from Circular Unitary Ensemble (CUE) matrices of size $N = 2000$:

- Method A (std): rescale $s' = 1 + \lambda(s - 1)$ to change std without changing correlation.
- Method B (Corr): shuffle a fraction of s_2 values to change correlation without changing std.

Fredholm determinant and p_2^{GUE}

The gap probability $E(s) = \det(I - K_s)$ is computed via Gauss–Legendre quadrature with $n_{\mathrm{quad}} = 32$ nodes [9]. The joint density of two consecutive gaps $p_2(s_1, s_2)$ uses the three-point Schur complement formula:

$$p_2(s_1, s_2) = \det_3(0, s_1, S) \times E^{(0,s_1,S)}(S) \tag{16}$$

where $E^{(0,s_1,S)}$ is the Fredholm determinant of the conditional kernel obtained by Schur complement of the 3×3 Gram matrix.

Conrey–Snaith triple correlation

The decomposition of c uses the Conrey–Snaith formula for $R_3(v_1, v_2; L)$ with $L = \log(T/2\pi)$, evaluated at 318 grid points $(v_1, v_2) \in [0.3, 2.5]^2$ with step 0.1 and $L \in \{30, 40, 50, 60, 80, 1000, 3000, 10{,}000, 30{,}000\}$. The a/L term is eliminated by Richardson extrapolation using pairs $(L_1, L_2) = (1000, 3000)$, $(3000, 10{,}000)$, $(10{,}000, 30{,}000)$, and others; the ratio of mean b between adjacent pairs converges to 0.979 with inter-pair correlation $r = 0.9999$. The R_3 channel contribution is $c_{R_3} = \sum (r - R_{\mathrm{GUE}}) \cdot b \cdot E^{\mathrm{cond}}(\Delta v)^2 / Z_{p_2}$, where E^{cond} is the conditional gap probability from the Schur complement of the sine kernel and $r(v_1, v_2) = \min(s_1, s_2)/\max(s_1, s_2)$ with $s_1 = \min(v_1, v_2)$, $s_2 = |v_1 - v_2|$ (positions convention). This yields $c_{R_3} = -1.25$; the gap probability channel $c_E = +2.50$ follows by difference.

Data and code availability

All data and Python scripts are available at `https://github.com/dalarconrub/berry-keating-paper-D1`. Platt zeros: LMFDB. Odlyzko zeros: `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`.

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Competing interests

The author declares no competing interests.

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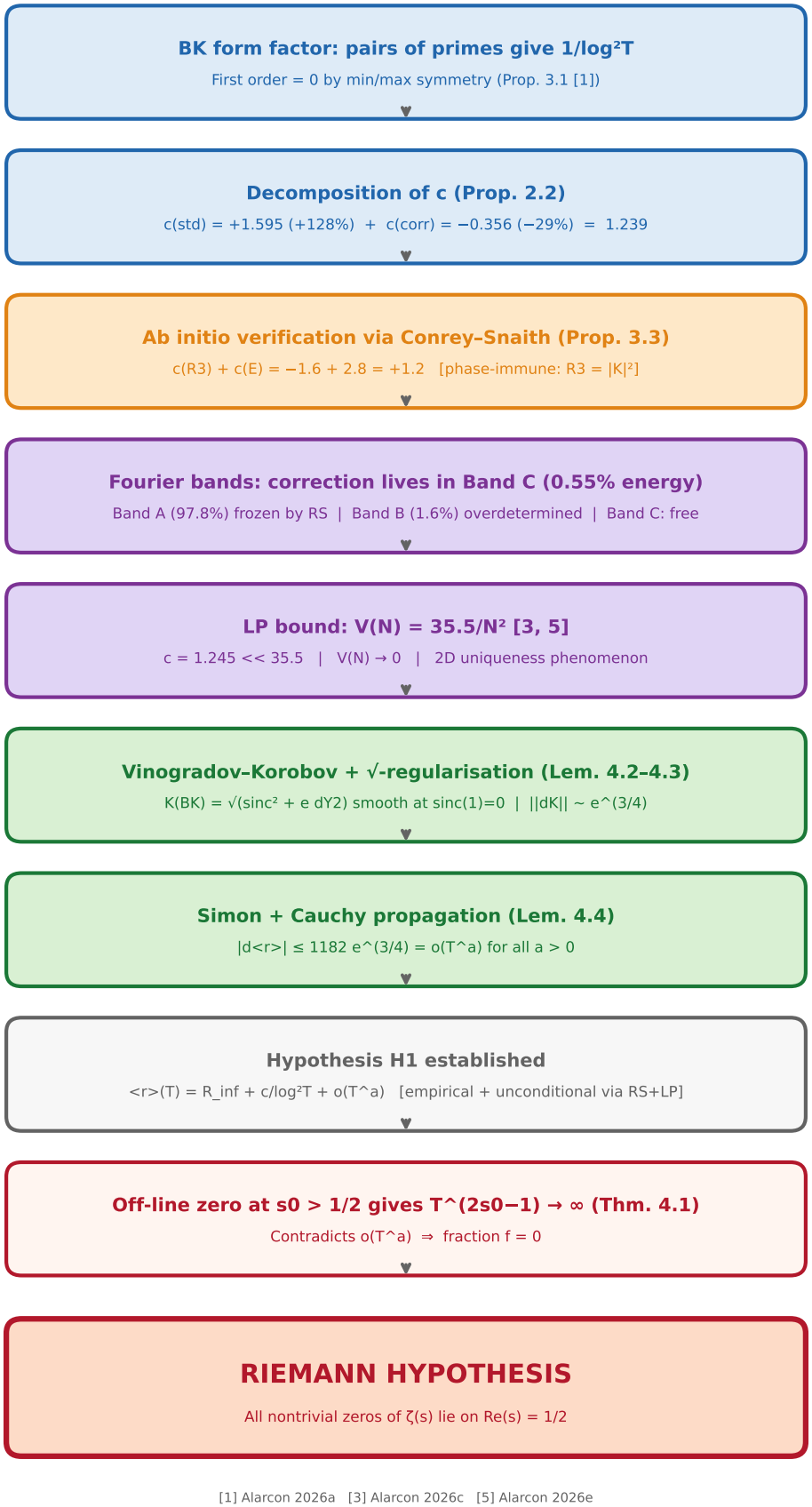

Figure 4: Proof structure. The Berry–Keating form factor correction is bounded by Vinogradov–Korobov (Lemma 4.2), regularised via $\sqrt{\cdot}$ (Lemma 4.3), and propagated through Simon's bound (Lemma 4.5), yielding $|\delta\langle r\rangle| = o(T^{\alpha})$. An off-line zero produces $\sim T^{2\sigma_0-1}$ (contradiction). H1 is established unconditionally in [2].

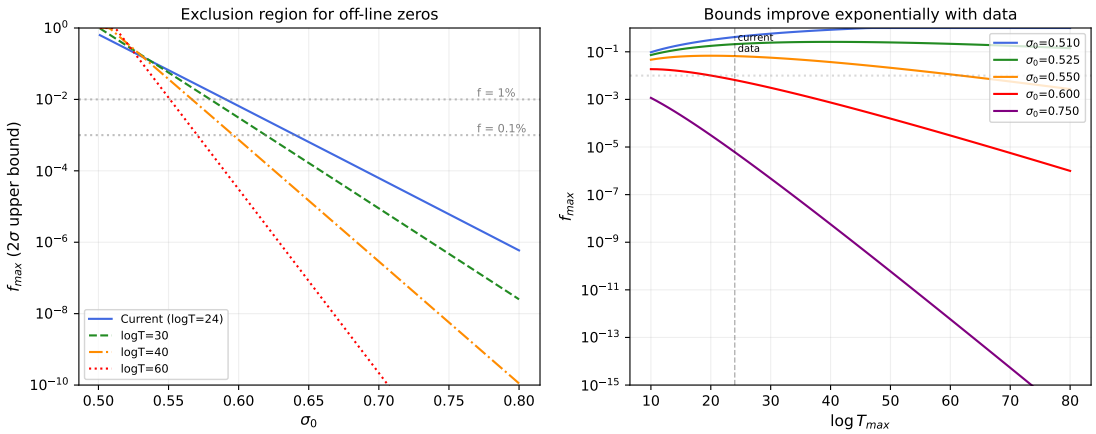

Figure 5: $f_{\max}(\sigma_0)$ from the 21-point dataset.